

# Qatar-1b

R-1911

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_131225 · created 2026-08-05T23:37:58+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456651.58998 +/- 0.00019 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.168 +/- 0.034                   |
| Transit depth (Rp/Rs)^2                           | 2.83 +/- 1.14 [%]                 |
| Orbital Inclination (inc)                         | 81.86 +/- 1.39                    |
| Airmass coefficient 1 (a1)                        | 1.022 +/- 0.0091                  |
| Airmass coefficient 2 (a2)                        | -0.011 +/- 0.011                  |
| Scatter in the residuals of the lightcurve fit is | 0.89 %                            |
| Best Comparison Star                              | #2 - [449, 333]                   |
| Optimal Aperture                                  | 3.19                              |
| Optimal Annulus                                   | 14.65                             |
| Transit Duration (day)                            | 0.0376 +/- 0.0092                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.168 ± 0.034 vs 0.1463 (deviation 0.64σ) |
| Duration fit vs published | 0.0376 d vs 0.0692 d (deviation 3.43σ)    |
| Mid-time O–C              | -0.5 min                                  |
| Depth SNR                 | 4.9                                       |
| Residual scatter          | 0.89 %                                    |

### Light curve

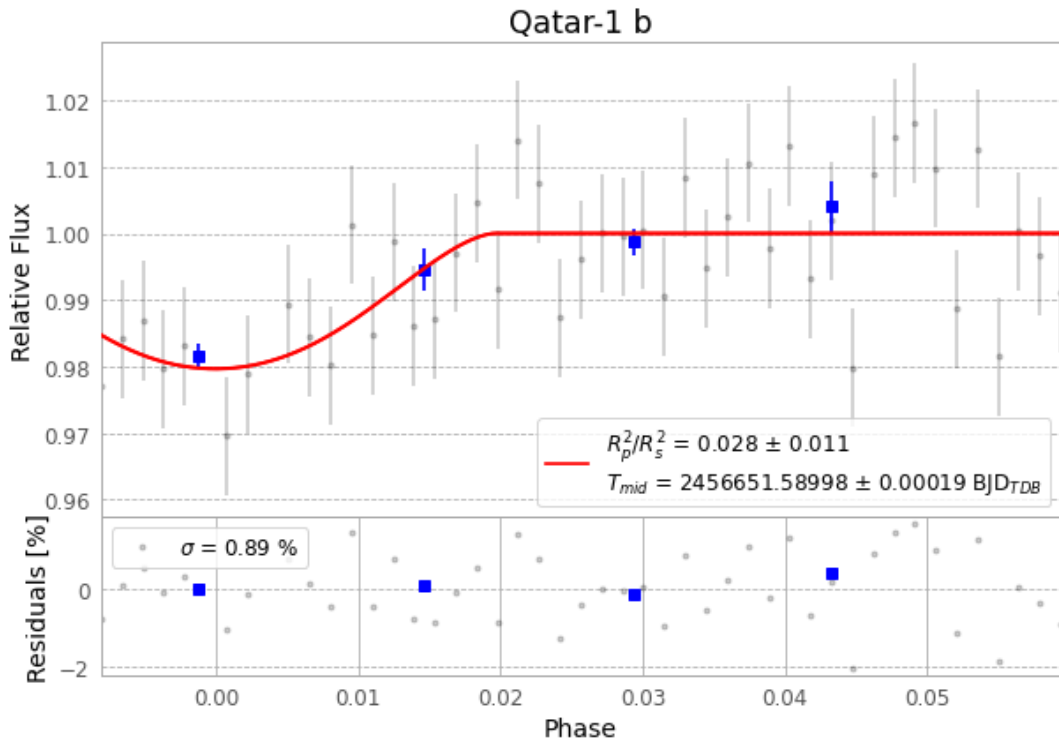


Figure 1 — FinalLightCurve\_Qatar-1 b\_2013-12-24.png (SHA-256 58df708930a4439e...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 257], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 43e885ae4635dabb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

47 FITS frames (31 MB), Qatar-1131225015112.FITS → Qatar-1131225040912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-131225.log (90ea2475765e...), FinalLightCurve\_Qatar-1 b\_2013-12-24.png (58df708930a4...), FinalLightCurve\_Qatar-1 b\_2013-12-24.csv (dd90ecd34515...)

## Compute resources

Wall time 142.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)